Statistical Report

Determinants of Belgian Household Energy Consumption

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1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, electricity consumption provides a useful proxy for household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants.

Dwelling-related attributes such as floor area, dwelling type, construction period, and Energy Performance Certificate (EPC) labels are consistently identified as important predictors of electricity demand (McLoughlin et al., 2013; Santin et al., 2009). Intuitively, larger dwellings typically require more energy for lighting, heating, cooling, and appliance use, while older buildings are constructed with materials which provides poorer insulation and therefore larger heat dissipation. Nevertheless, these variables interacts with socio-economic factors. Larger households topically host a higher number of occupants, therefore they can share the consumption for heating, refrigerators and air conditioning lowering the per-capita consumption. Similarly, high EPC label buildings are prerogative of high-income classes which might use a larger number of appliances and indulge into energy-consuming behaviours going to erode the benefits of a properly insulated house in terms of energy savings (Aydin et al., 2017).

Using household electricity consumption as a proxy for energy demand, this cross-sectional study examines its relationship with building characteristics (e.g. floor area, EPC label, construction era) and socio-economic factors (number of permanent residents and annual net household income) collecting data from 2500 households in Belgium. Importantly, the study aims at characterising the interplay between EPC labels and the other factors involved in energy consumptions to provide data-driven tools to decision-makers.

2 Data management

2.1 Data Cleaning

After loading the data, three observations (rows 66, 36, and 135) were removed from the dataset according to the instructions. Subsequently, the dataset was cleaned by excluding observations with missing values in the variables EPC label (`epc_label`) and annual household electricity consumption (`annual_kwh`). This step ensures that the analysis is conducted only on complete cases for the key variables of interest. None of the variables displayed missingness above 5%. Missing values in other variables were retained (see appendix) without imputation. To inspect the distribution of electricity consumption and identify potential extreme values, the observations were sorted in decreasing order of annual electricity consumption (`annual_kwh`). Five observations (row 267, 1330, 1570, 369, 12) were two orders of magnitude larger than the others (172 to 312 times larger) and, therefore, were removed as were attributed to imputation error.

2.2 Data Coding

Variables were coded according to their type: continuous (e.g., `annual_kwh`, `sq_meters`, `income_euro`), count (`occupancy_count`), ordinal (`epc_label`, `building_era`), categorical (`roof_color`), or binary (`pet_ownership`). Table 1 summarizes their descriptions and coding schemes.

Table 1: Data Dictionary

Name	Description	Type_Coding
annual_kwh	yearly electricity use	Continuous (kWh/year)
sq_meters	indoor living area	Continuous (m^2)
epc_label	energy performance certificate	Ordinal: A, B, ..., F
building_era	construction period	Ordinal: dummy coded; ref = Pre-1970
income_euro	net household income	Continuous (€/year)
occupancy_count	permanent residents	Count: 1, 2, ..., n
dist_to_brussels	distance to Brussels center	Continuous (km)
roof_color	roof tile color	Categorical: e.g., red, black, other
pet_ownership	owner has a pet	Binary: 0 = No, 1 = Yes

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels is relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E occur less frequently, 13.6% and F 9.6% respectively. Buildings constructed between 1970 and 2000 represent 34.4% of the sample, pre-1970 buildings account for 33.1%, and post-2000 buildings make up 32.6%. Occupancy count ranges from 1 to 7 with all counts accounting approximately 14-15%. Red roofs are the most common (35.2%), followed closely by black roofs (33.3%) and other colours (31.5%). Almost one in four (39.6%) owners has a pet, while 60.4% does not. None of the categories appear too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. Overall, the categorical variables show meaningful variation across categories. From the exploratory analysis, two categorical variables seem to correlate strongly with electricity consumption: EPC label and occupancy count. The other categorical variables do njot display an apparent correlation with electricity consumption. Summary tables and plots are presented in the appendix.

3.2 Continuous variables

On average, households consume 5,874.76 kWh of electricity annually, with a relatively large standard deviation (2,224.25 kWh). The median consumption (5,432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also supported by the QQ plot showing upward deviations above the diagonal line in the upper tail (see appendix). Floor area averages

158.25 m² with a standard deviation of 36.50 m², and a range from 60.26 m² to 341.30 m². The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels is 62.96 km, with observations ranging from 5.11 km to 120 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income averages €48,842.90, with a standard deviation of €22,534.97 and a range from €8,487.91 to €206,367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area (r = 0.38) and with income (r = 0.29), suggesting that larger and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive correlation with each other, namely floor area and income (r = 0.72). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not strongly influence household energy use, dwelling size, or income in this dataset.

Summary tables and graphs are provided in the appendix.

4 Inferential Statistics

4.1 Regression model

4.2 DAGs

4.3 Results table

Term	Estimate	Std. Error	t value	Pr(> t

- VIF analysis
- Cross validation analysis

4.4 Diagnostic plots

- Residuals vs Fitted
- Normal Q–Q Plot
- Scale–Location Plot (Spread–Location)
- Residuals vs Leverage
- Influence (Cook's Distance)

4.5 Contrast analysis

- for research question 2 and 4

5 Conclusions

6 Appendix

6.1 Summary tables for non-continuous varibales

Table 3: EPC Label Distribution

epc_label	n	percent
A	552	22.2
B	436	17.6
C	529	21.3
D	389	15.7
E	337	13.6
F	239	9.6

Table 4: Building Era Distribution

building_era	n	percent
1970-2000	853	34.4
Post-2000	808	32.6
Pre-1970	821	33.1

Table 5: Roof Color Distribution

roof_color	n	percent
Black	826	33.3
Other	783	31.5
Red	873	35.2

Table 6: Pet Ownership Distribution

pet_ownership	n	percent
0	1500	60.4
1	982	39.6

Table 7: Occupancy Count Distribution

occupancy_count	n	percent
1	350	14.1

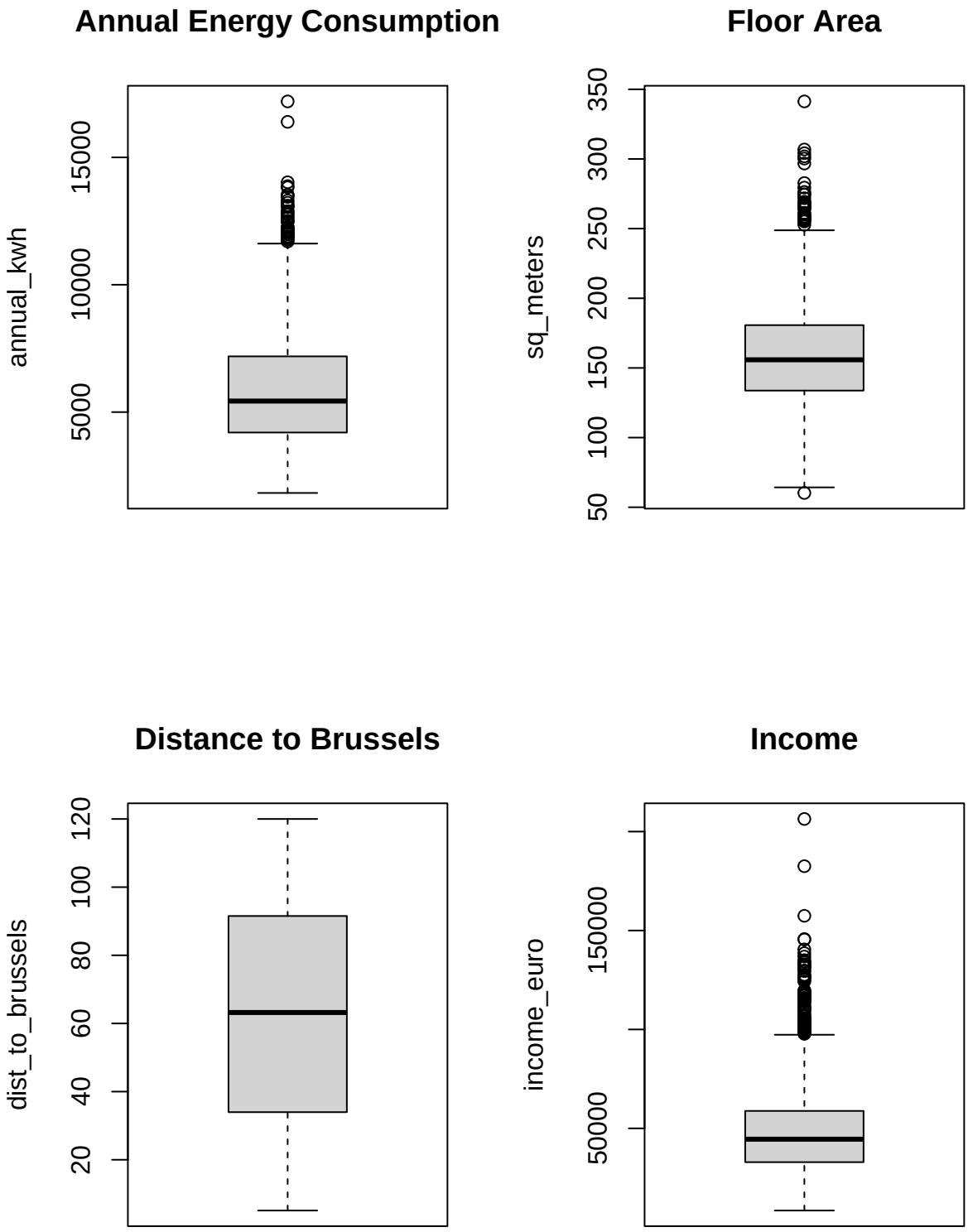
2	354	14.3
3	342	13.8
4	376	15.1
5	375	15.1
6	341	13.7
7	339	13.7
NA	5	0.2

6.2 Summary tables for continuous varibales

Table 8: Summary Statistics for Continuous Variables

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	5874.76	158.25	62.96	48842.90
SD	2224.25	36.50	33.04	22534.97
Median	5432.07	155.86	63.21	44535.07
IQR	2987.79	46.95	57.55	25894.30
Min	1824.63	60.26	5.11	8487.91
Max	17197.12	341.30	120.00	206367.91

6.3 Summary plots for continuous varibales



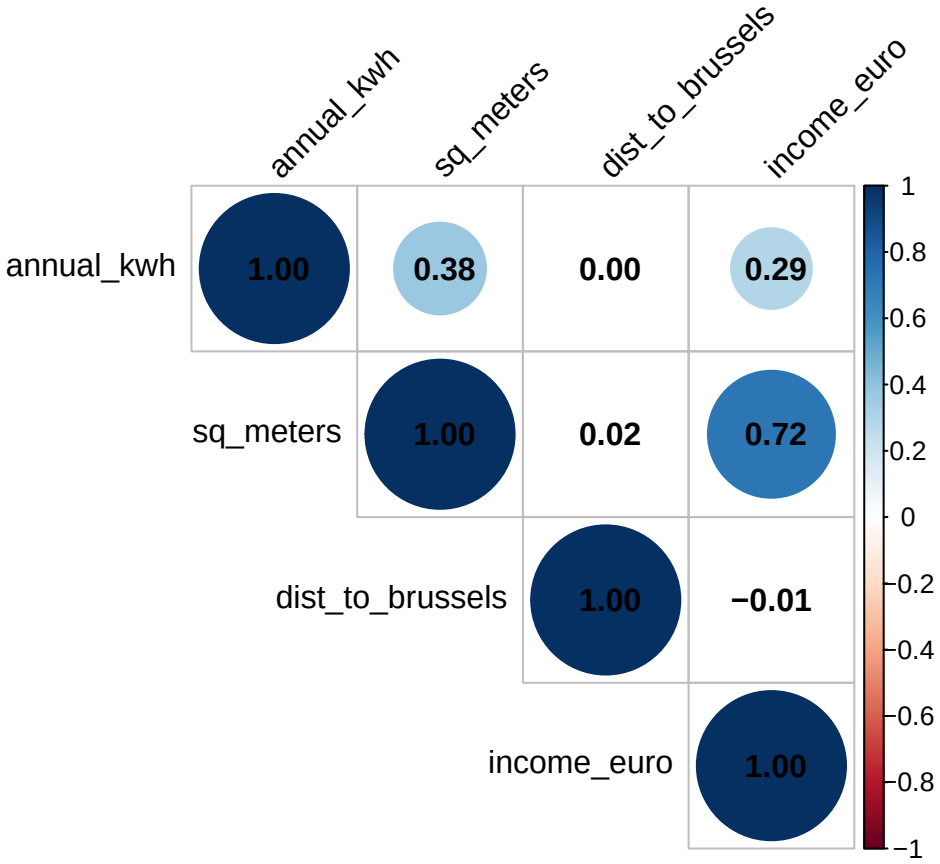
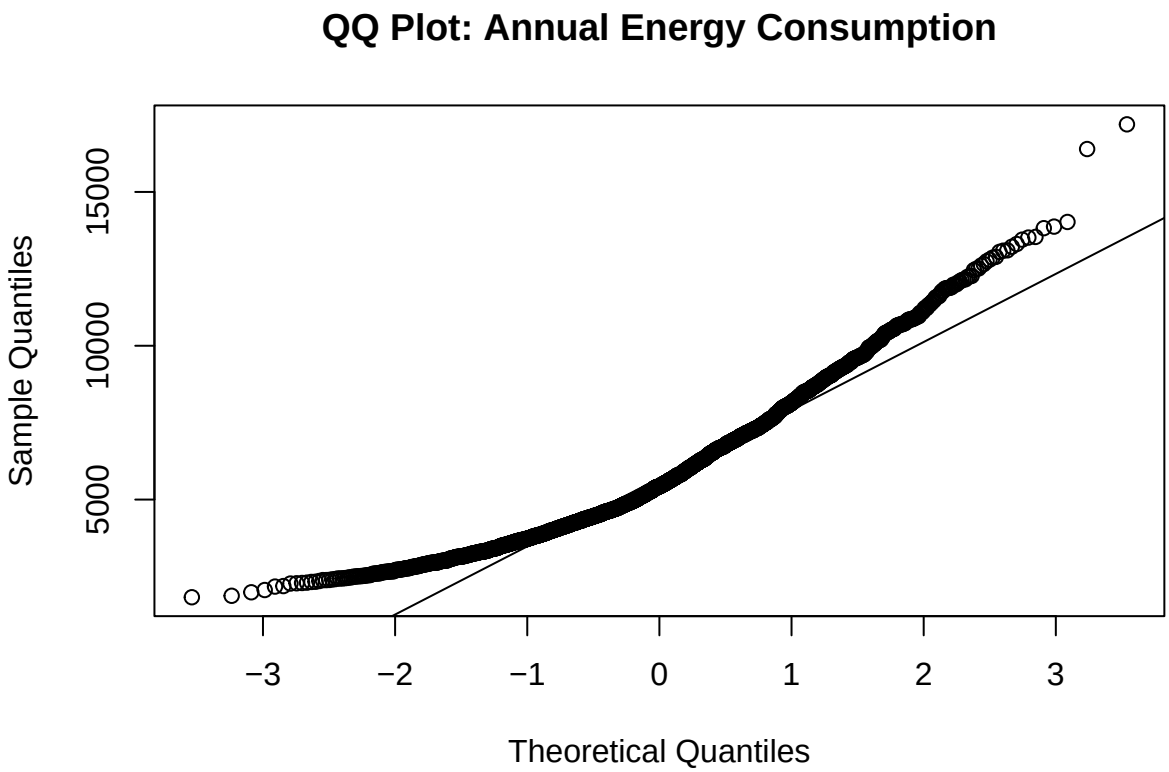
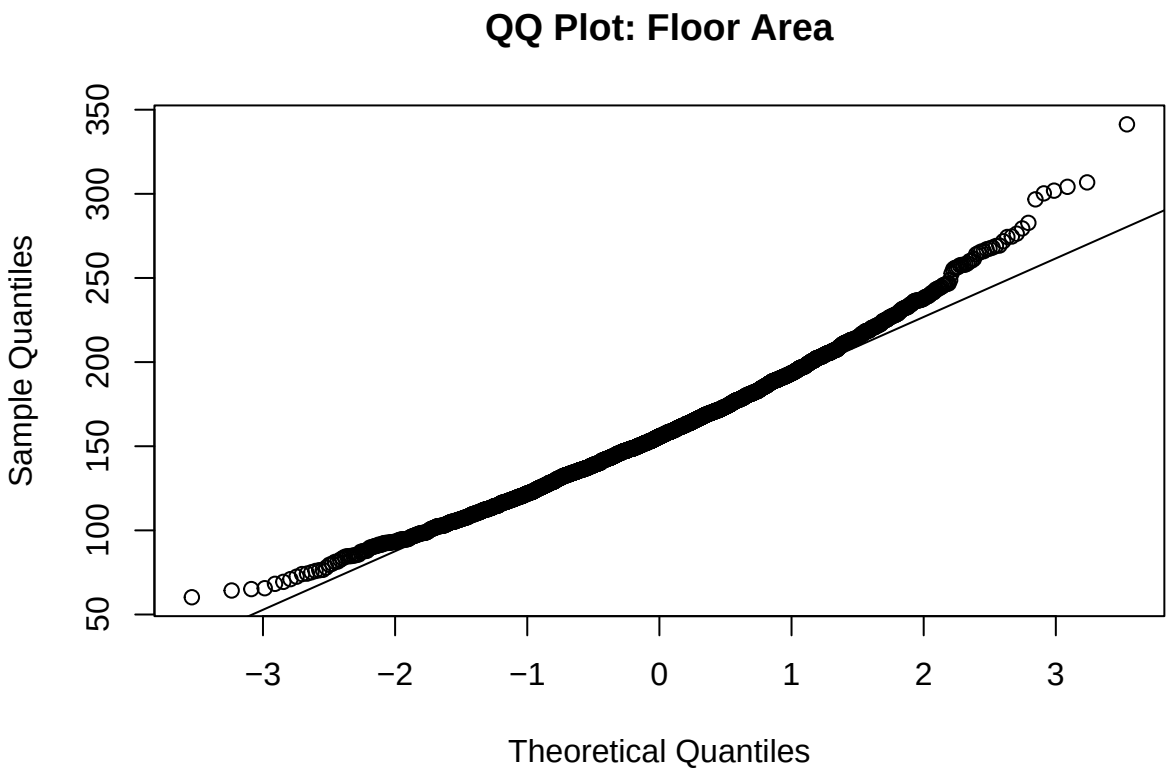
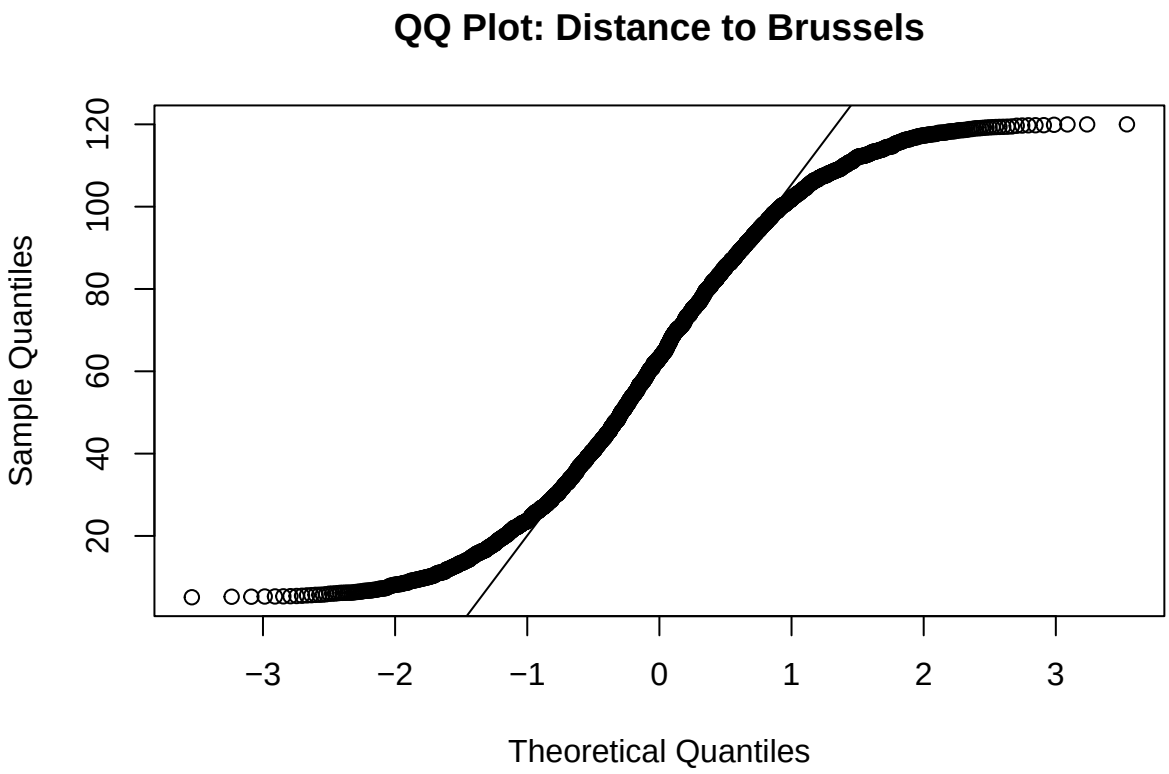
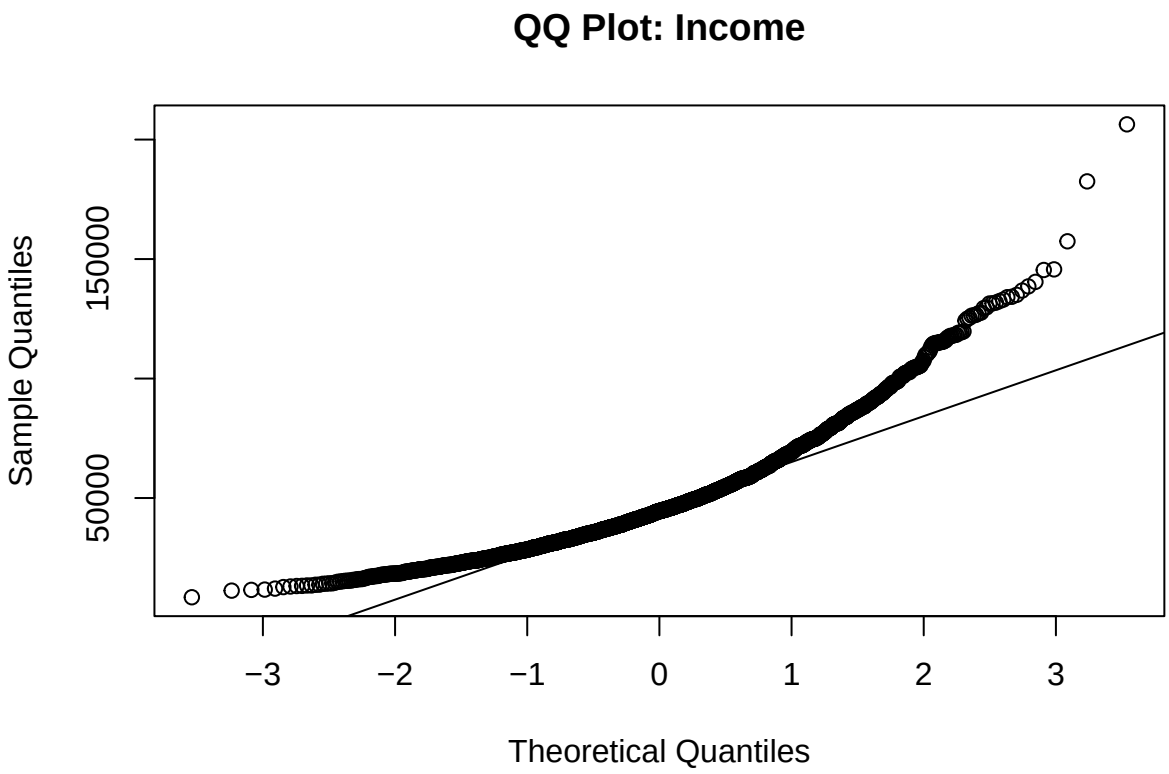


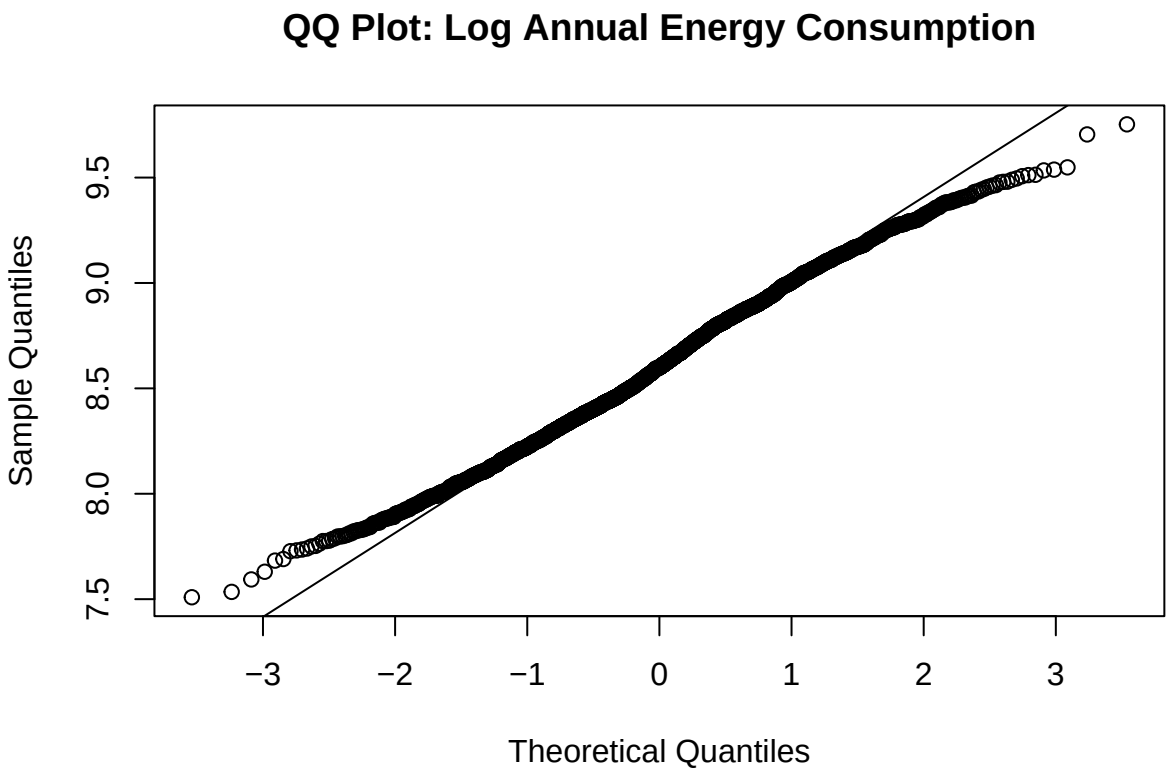
Figure 1: Correlation plots for continuous variables

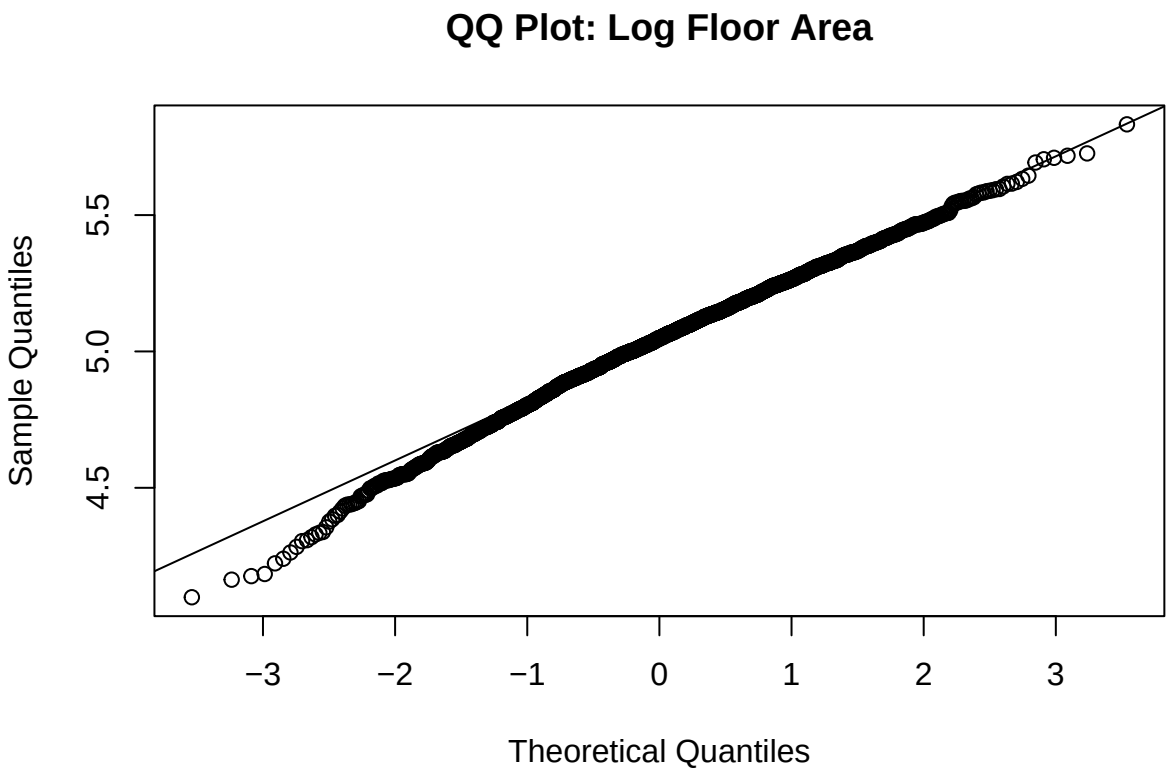

QQ Plot: Annual Energy Consumption
Sample Quantiles
5000
10000
15000
−3
−2
−1
0
1
2
3
Theoretical Quantiles

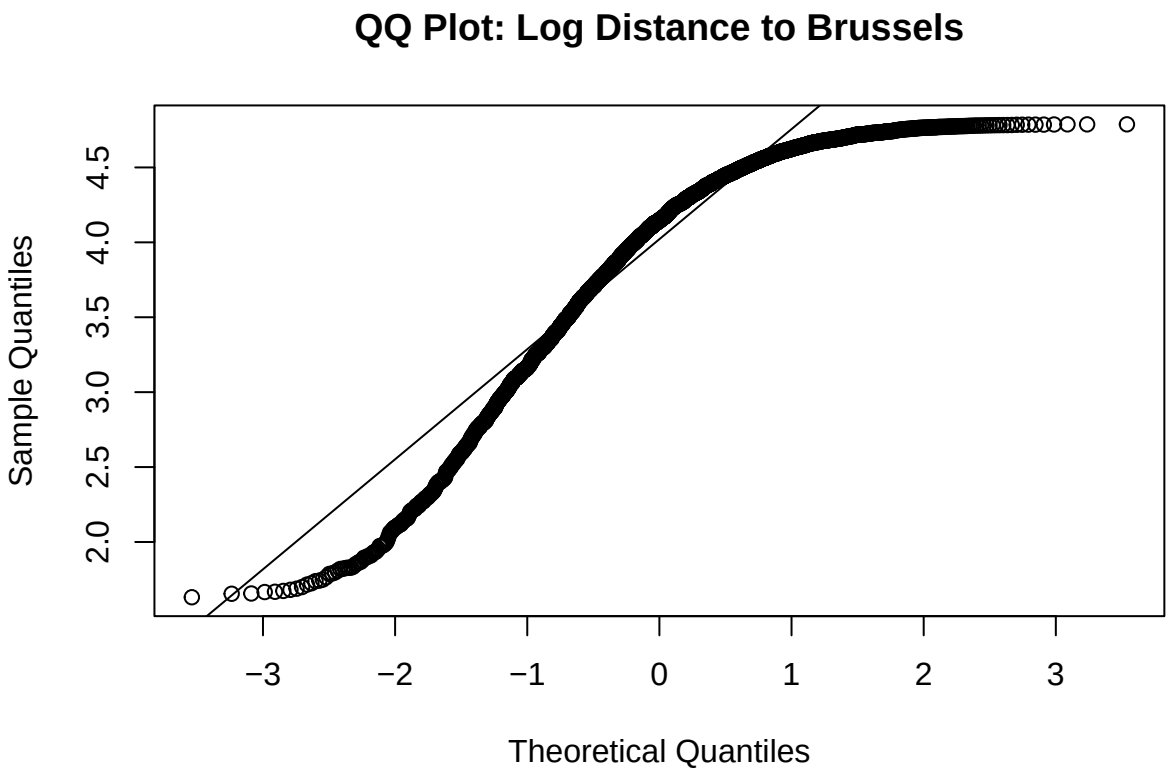

QQ Plot: Floor Area
Sample Quantiles
50
100
150
200
250
300
350
−3
−2
−1
0
1
2
3
Theoretical Quantiles

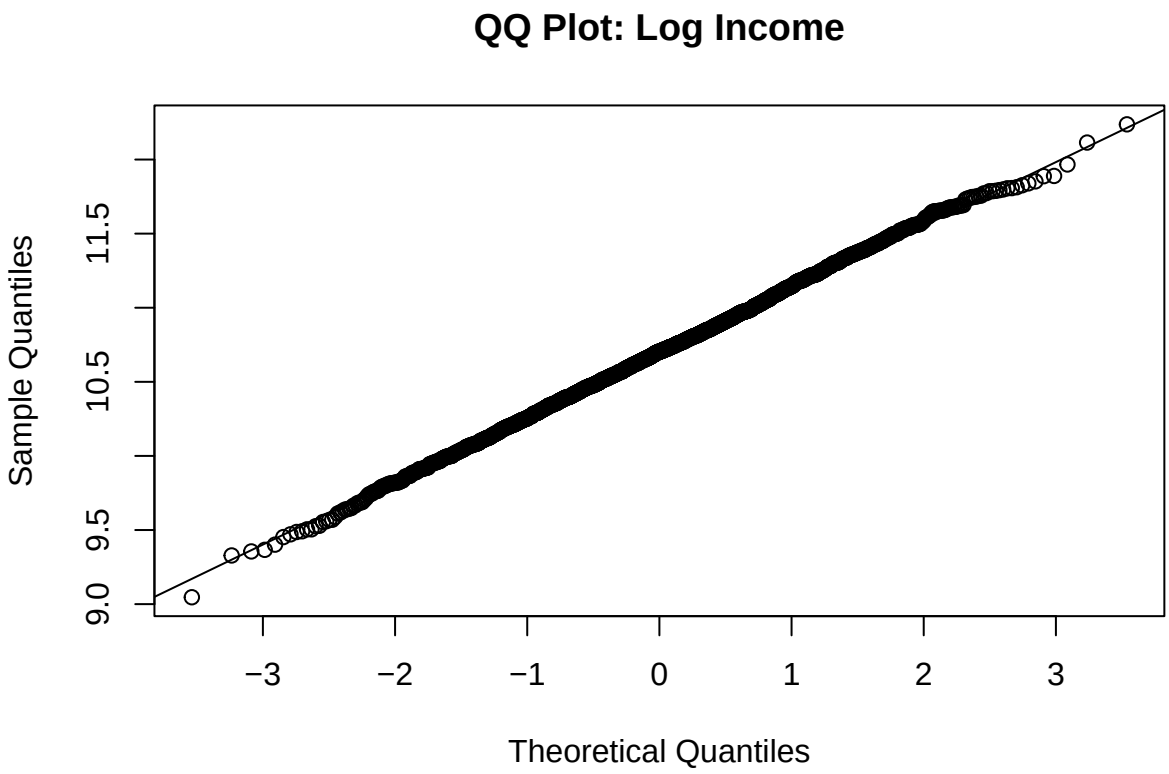

QQ Plot: Distance to Brussels
Sample Quantiles
20
40
60
80
100
120
−3
−2
−1
0
1
2
3
Theoretical Quantiles

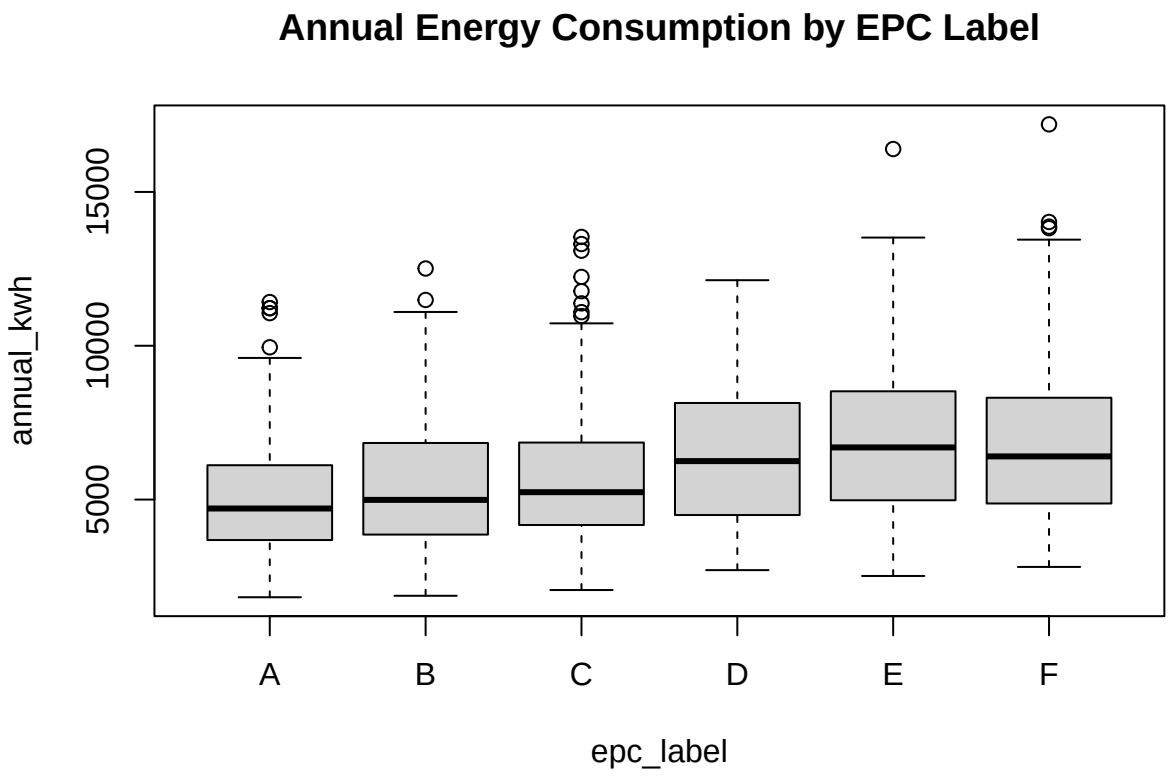

QQ Plot: Income
Sample Quantiles
50000
150000
−3
−2
−1
0
1
2
3
Theoretical Quantiles

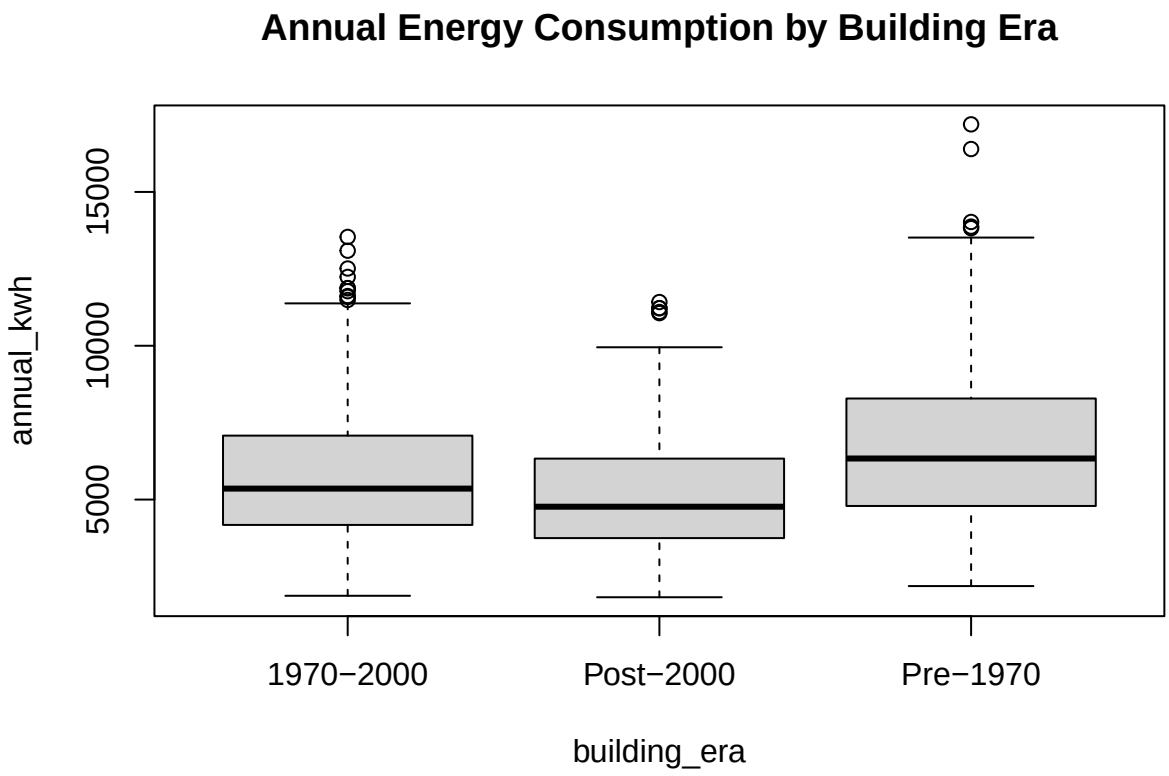

QQ Plot: Log Annual Energy Consumption
Sample Quantiles
7.5
8.0
8.5
9.0
9.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles

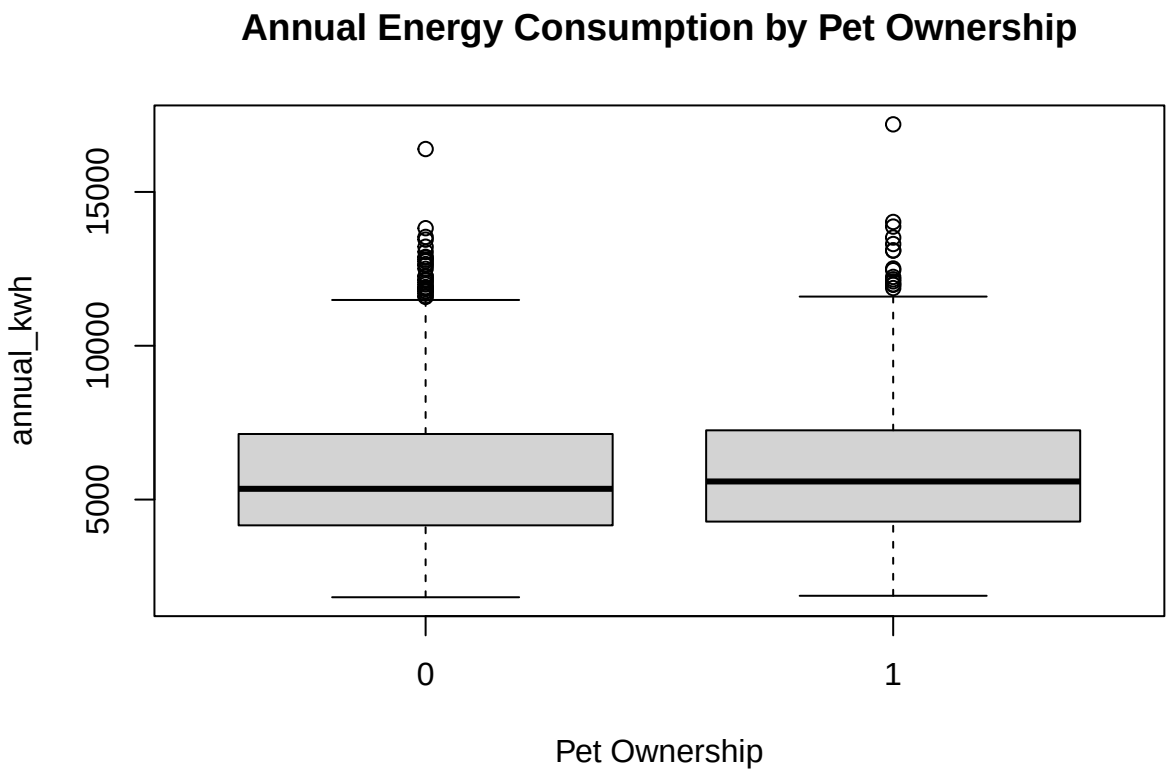

QQ Plot: Log Floor Area
Sample Quantiles
4.5
5.0
5.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles

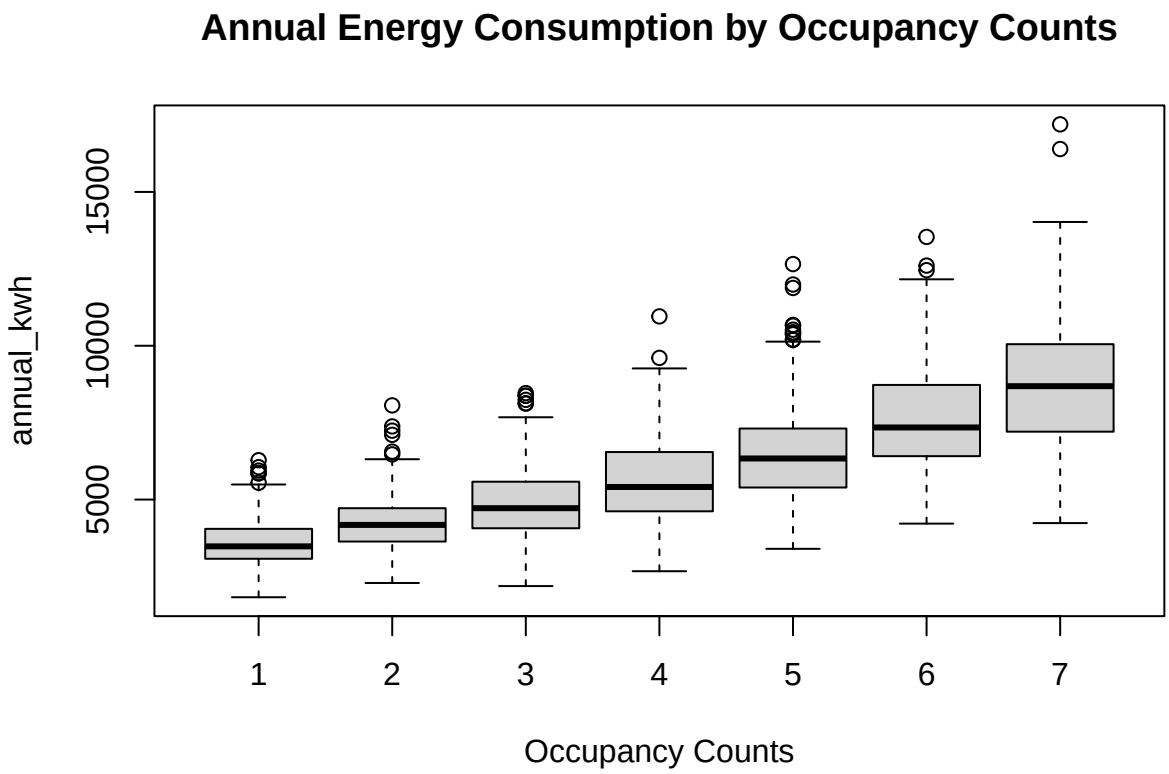

QQ Plot: Log Distance to Brussels
Sample Quantiles
2.0
2.5
3.0
3.5
4.0
4.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles

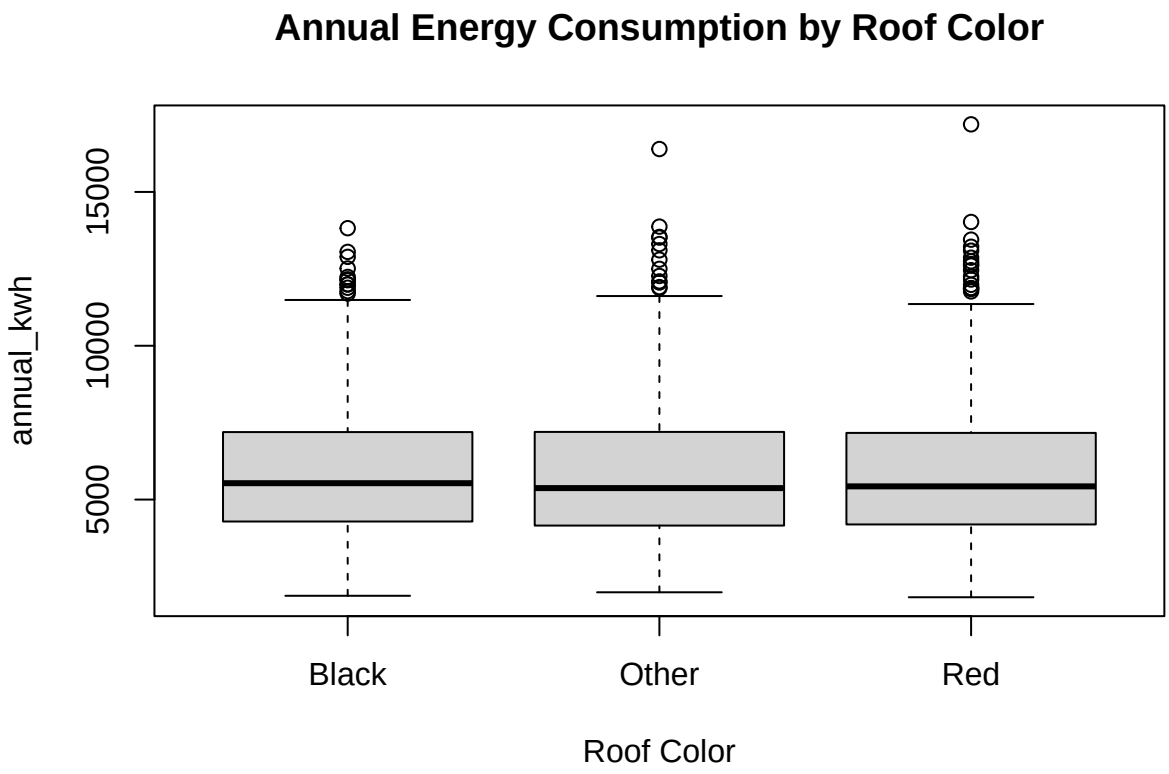

QQ Plot: Log Income
Sample Quantiles
9.0
9.5
10.5
11.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Annual Energy Consumption by EPC Label
annual_kwh
5000
10000
15000
A
B
C
D
E
F
epc_label


Annual Energy Consumption by Building Era
annual_kwh
5000
10000
15000
1970−2000
Post−2000
Pre−1970
building_era


Annual Energy Consumption by Pet Ownership
annual_kwh
5000
10000
15000
0
1
Pet Ownership


Annual Energy Consumption by Occupancy Counts
annual_kwh
5000
10000
15000
1
2
3
4
5
6
7
Occupancy Counts


Annual Energy Consumption by Roof Color
annual_kwh
5000
10000
15000
Black
Other
Red
Roof Color

References

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McLoughlin, F., Duffy, A., & Conlon, M. (2013). Determinants of residential electricity consumption: Using smart meter data to examine the effect of climate, building characteristics, appliance stock, and occupants' behavior. *Energy*, *55*, 184–194.

Santin, O. G., Itard, L., & Visscher, H. (2009). The effect of occupancy and building characteristics on energy use for space and water heating in dutch residential stock. *Energy and Buildings*, *41*(11), 1223–1232.